



Intro to HTML Structure

Unit 1 - Lesson 2: The Skeleton of Every Web Page!



Open VS Code → Create index.html → Let's build!

What We'll Learn Today

- ✓ Identify the essential parts of an HTML document
- ✓ Explain what each HTML tag does in the basic structure
- ✓ Create a properly formatted HTML file from scratch
- ✓ Understand the difference between tags, elements, and attributes

By the end: You'll write your first HTML page! 



What Is HTML?

HTML = HyperText Markup Language



A markup language — NOT a programming language



It **annotates** text to make it distinct from content



Think of HTML as the **skeleton** of a web page



A markup language is simply a way of annotating a document — like a teacher marking up a paper with notes!



The Three Pillars of Web Development

Technology	Role	Analogy
HTML	 Structure	The skeleton of a house
CSS	 Style	Paint, furniture, decorations
JavaScript	 Behavior	Electricity and plumbing

HTML is the foundation — without it, nothing else matters!



Tags vs. Elements

Tags are just the code:

.....

```
<p>  
</p>
```

Elements = Tag + Content:

.....

```
<p>This is a paragraph.</p>
```

Anatomy of an Element:

.....

```
<p>      This is a paragraph.      </p>  
  ↑                ↑                ↑  
Opening Tag      Content      Closing Tag
```

Most elements have an opening tag, content, and a closing tag!



The HTML Skeleton (Boilerplate)

```
<?xml version="1.0" encoding="UTF-8" standalone="no" ?>
<?xml-stylesheet type="text/css" href="http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd" ?>
<html>
  <head>
    <meta charset="UTF-8" />
    <title>
    </title>
  </head>
  <body>
    <!-- Everything visible goes here -->
  </body>
</html>
```

This is the standard starting point for every HTML file. Memorize it!

The 4 Essential Parts

1 `<!DOCTYPE html>`

Tells the browser this is **HTML5**. Must be the first line!

2 `<html>`

The **root element** that wraps everything on the page.

3 `<head>`

Behind-the-scenes information (metadata). Not visible to users.

4 `<body>`

All **visible** content on the page (text, images, links, etc.).

The `<head>` Section

-  Contains information ABOUT your webpage
-  Users do **NOT** see the head content on the page
-  Holds: Title, meta info, CSS links, JS links

Common Head Elements:

Element	Purpose
<code><title></code>	Shows in browser tab
<code><meta charset></code>	Character encoding (UTF-8)
<code><meta viewport></code>	Mobile responsiveness
<code><link></code>	CSS stylesheets
<code><script></code>	JavaScript files

The `<body>` Section

-  Contains **ALL visible content**
-  Users see everything in the body
-  Includes: Headings, paragraphs, images, links, lists, forms, buttons

Visible Content Elements:

-  Headings (`<h1>` - `<h6>`)
-  Paragraphs (`<p>`)
-  Images (`<img>`)
-  Links (`<a>`)
-  Lists (`<ul>` , `<ol>` , `<li>`)
-  Forms (`<form>` , `<input>` , `<button>`)

If it's not in the `<body>` — the user won't see it!



Visualizing the Structure

The HTML Skeleton (Annotated)

`<!DOCTYPE html>` ← 1. Declaration (not a tag!) →

`<html lang="en">` ← 2. Root element →

`<head>` ← 3. Metadata goes here →

`<title>My Page</title>`

`</head>`

`<body>` ← 4. Visible content goes here →

`<h1>Hello, World!</h1>`

⚡ Emmet Shortcut: Instant Skeleton



Type ! then press TAB in VS Code

VS Code instantly generates the entire HTML skeleton for you!

VS Code Emmet is a plugin for text editors that significantly speeds up HTML/CSS workflow by using abbreviated syntax that expands.



Let's Code Together!

Open VS Code → Create index.html → Type ! + TAB!

```
14 ben een front-end developer en student webdevelopment
15
16 </p>
17
18 <hrer= #work--section" class="pink--button scroll">Mijn Skills
19 </article>
20 
22 <section id="skills--section">
23   <h1 class="section--header">
24     Mijn Skills.
25   </h1>
26   <section id="skills--section--wrap">
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
```

Step 1: Create Your First HTML File

Open VS Code

- 1 Go to **File** → **Open Folder** → Create a new folder called `my-first-page`

Create a new file

- 2 Right-click on the folder → **New File** → Name it `index.html`

Type `!` and press TAB

- 3 Emmet generates the complete skeleton for you!



You now have a valid HTML file! Let's add content!



Step 2: Add Content to `<body>`

```
<span class="html-bracket">&lt;/span><span class="html-tag">body</span><span class="html-bracket">&gt;</span>
<span class="html-bracket">&lt;/span><span class="html-tag">h1</span><span class="html-bracket">&gt;</span>My First Web Page<span class="html-bracket">&lt;/span><span class="html-tag">h1</span><span class="html-bracket">&gt;</span>
<span class="html-bracket">&lt;/span><span class="html-tag">p</span><span class="html-bracket">&gt;</span>Welcome to my awesome website!<span class="html-bracket">&lt;/span><span class="html-tag">p</span><span class="html-bracket">&gt;</span>
<span class="html-bracket">&lt;/span><span class="html-tag">img</span><span class="html-bracket">&gt;</span> <span class="html-attr">src</span><span class="html-bracket">&lt;/span><span class="html-content">"https://placeholder.co/400x200"</span> <span class="html-attr">alt</span><span class="html-bracket">&lt;/span><span class="html-content">"A placeholder image"</span><span class="html-bracket">&gt;</span>
<span class="html-bracket">&lt;/span><span class="html-tag">body</span><span class="html-bracket">&gt;</span>
```

💡 Notice: `<img>` is a **self-closing** tag — no closing tag needed!

Save → Open in browser → See your page! 



The Browser's Job

The Browser:

- Reads HTML documents
- Understands the structure
- Renders the page visually
- Applies the skeleton you created

Common Browsers:

- ● Chrome (most popular)
- ● Edge (Windows)
- 🦊 Firefox (Mozilla)
- 🍏 Safari (Mac)

Your browser = The HTML interpreter!

The purpose of a web browser is to read HTML documents and display them correctly.



Understanding HTML Syntax

Tag Anatomy Breakdown



```
<span class="html-bracket">&lt;</span><span class="html-tag">p</span> <span class="html-attr">class</span><span class="html-bracket">=</span><span class="html-content">"intro"</span><span class="html-bracket">&gt;</span>Hello <span class="html-tag">world</span>!<span class="html-bracket">&lt;</span><span class="html-tag">p</span><span class="html-bracket">&gt;</span>
```

< = Opening bracket

p = Tag name (paragraph)

class = Attribute name

"intro" = Attribute value

Hello world! = Content

</p> = Closing tag

Attribute: Extra information about a tag (e.g., **class**, **id**, **src**, **href**)



5 Common HTML Mistakes

✗ Forgetting closing tags

Every `<html>` needs `</html>`

✗ Wrong order

`<head>` must come before `<body>`

✗ Missing DOCTYPE

Always start with `<!DOCTYPE html>`

✗ Wrong file extension

Must end in `.html`, not `.txt`

✗ Typos in tag names

`<html>` won't work — it's `<html>`!

**All of these break your page or
make it render incorrectly!**



Check Your Understanding

1. What does `DOCTYPE` tell the browser?
2. Which tag wraps ALL content on a webpage?
3. Where does the visible content live — `<head>` or `<body>`?
4. What is the difference between a **tag** and an **element**?
5. What does the Emmet shortcut `!` + `TAB` do in VS Code?

✓ Check Your Understanding (Answers)

1. `DOCTYPE` tells the browser that the document is HTML5.
2. The `<html>` element is the **root element** — it contains everything.
3. The `<body>` contains all visible content. The `<head>` is metadata only.
4. A **tag** is just the code (`<p>`). An **element** is the tag + content (`<p>Hello</p>`).
5. It instantly generates the full HTML skeleton (boilerplate) for you!

Key Takeaways

✓ HTML = Structure — the skeleton of every web page

✓ Tags are code; Elements = Tag + Content

✓ 4 Essential Parts: DOCTYPE, `<html>`, `<head>`, `<body>`

✓ `<head>` = Metadata (not visible); `<body>` = All visible content

✓ Emmet shortcut `!` + `TAB` = Instant skeleton



Quick Reference

Task	Code
Declare HTML5	<code><!DOCTYPE html></code>
Root element	<code><html lang="en">...</html></code>
Head section	<code><head>...</head></code>
Body section	<code><body>...</body></code>
Page title (tab)	<code><title>My Page</title></code>
Main heading	<code><h1>Heading</h1></code>
Paragraph	<code><p>Text</p></code>
Image	<code></code>
Emmet shortcut	<code>! + TAB</code>

What's Next?

Common HTML Elements!

 Headings, paragraphs, links, images, lists

 Understanding attributes like `href` and `src`

 Semantic HTML5 — building pages with meaning

Introduction to CSS — making it look beautiful! 



You Built Your First HTML Page!

Skeleton complete! 🦷 → Next step: Style & Structure! 🎨

Lesson 1.2 Complete! Keep coding! 💻

Keyboard Events lesson — dark navy background with white text and icons, white blocks with "Mac window" style shadows, dots, step boxes, and concept cards.